Experiment 6: Exploration of Cloud Service Models (IaaS, PaaS, SaaS) Using a Free-Tier Cloud Platform

Aim

To explore the cloud service models IaaS, PaaS and SaaS using a free-tier cloud platform.

Objective

- Understand the three cloud service models.
- Create a free-tier account and explore one service of each model.
- Compare the models based on control and responsibility.

Platform Used

Amazon Web Services (AWS) Free Tier. Google Cloud or Microsoft Azure free accounts can be used in the same way. Check the provider's website for the current free-tier limits.

Theory

Model	Meaning	User manages	Example
IaaS	Virtual servers, storage and networks on rent	OS, runtime, applications, data	AWS EC2, Google Compute Engine
PaaS	Ready platform to build and run applications	Application code and data	AWS Elastic Beanstalk, Google App Engine
SaaS	Complete software used through a browser	Only own data and settings	Gmail, Zoho CRM, Microsoft 365

Procedure

Part A - IaaS (virtual machine)

1. Sign up for an AWS free-tier account and sign in to the console.
2. Open the **EC2** service and click Launch Instance.
3. Choose a free-tier eligible Linux image and a t2.micro / t3.micro instance type.
4. Create a key pair, keep the default security group and launch.
5. Note the public IP and connect with SSH / EC2 Instance Connect.
6. **Stop or terminate the instance after the experiment to avoid charges.**

Part B - PaaS (deploy an application)

1. Open **Elastic Beanstalk** (or App Engine) and create a new web application.
2. Select a platform such as Python or Node.js and use the sample application.
3. Click Create and wait for the environment to become healthy; open the application URL.
4. Observe that servers, OS and scaling are handled by the platform.

Part C - SaaS (use a ready application)

1. Sign in to a SaaS tool such as Google Docs or Zoho Writer.
2. Create a document and share it with another user.
3. Observe that nothing needs to be installed or maintained by the user.

Attach screenshot here - EC2 instance running

Attach screenshot here - PaaS application URL / SaaS document

Comparison

Criteria	IaaS	PaaS	SaaS
Control	High	Medium	Low
Setup effort	High	Medium	Very low
Maintenance by user	OS, software, security patches	Code only	None
Best for	Admins, custom setups	Developers	End users
Scalability	Manual / configured	Automatic	Provider managed

Results

One service from each cloud model was explored. IaaS gave full control over a virtual machine, PaaS made deployment simple, and SaaS needed no setup at all.

Conclusion

The more the provider manages, the less control the user has but the faster the work is done. The right model depends on the skill of the team and the needs of the application.